WORKED SOLUTION

STEP III 2005, Question 5

Pure • Common Tangents to Parabolas

STEP III 2005, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q5

The key idea

Rather than labelling a tangent by its point of contact, label it by its gradient. A parabola has exactly one tangent of each gradient m, and the first part tells us its y-intercept. Two parabolas share a tangent of gradient m precisely when those two intercepts agree, so counting common tangents becomes counting the roots m of a single equation, which is a quadratic when $a_1 \neq a_2$ and linear when $a_1 = a_2$.

Solution

The tangent of gradient m

The gradient of $y = ax^2 + bx + c$ is $2ax + b$. Since $a \neq 0$, this equals m at exactly one point, $x_0 = \dfrac{m-b}{2a}$, with $y_0 = ax_0^2 + bx_0 + c$. The tangent there is $y - y_0 = m(x - x_0)$, that is $y - mx = y_0 - mx_0$. Using $m = 2ax_0 + b$,

$$\begin{aligned} y_0 - mx_0 &= ax_0^2 + bx_0 + c - (2ax_0 + b)x_0 \\ &= c - ax_0^2 = c - a \cdot \frac{(m-b)^2}{4a^2} = c - \frac{(m-b)^2}{4a}, \end{aligned}$$

so the tangent at P is $y - mx = c - \dfrac{(m-b)^2}{4a}$, as required.

The condition for a common tangent

By the first part, each curve has exactly one tangent of gradient m, namely

$$y = mx + k_1, \quad k_1 = c_1 - \frac{(m-b_1)^2}{4a_1}, \qquad \text{and} \qquad y = mx + k_2, \quad k_2 = c_2 - \frac{(m-b_2)^2}{4a_2}.$$

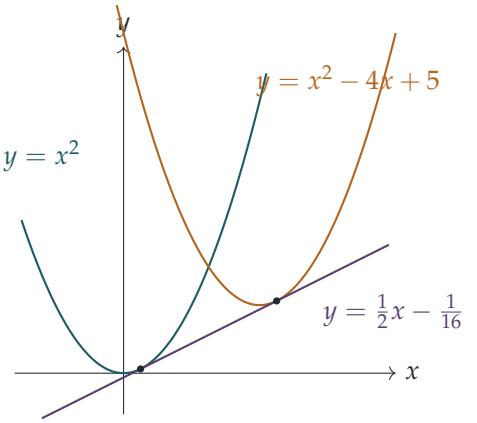
Two lines of the same gradient coincide exactly when their intercepts are equal, so the curves have a common tangent of gradient m if and only if $k_1 = k_2$. Multiplying by

$4a_1a_2 \neq 0$ (which doesn't change whether the equation holds),

$$\begin{aligned} 4a_1a_2(k_2 - k_1) &= 4a_1a_2(c_2 - c_1) - a_1(m - b_2)^2 + a_2(m - b_1)^2 \\ &= (a_2 - a_1)m^2 + 2(a_1b_2 - a_2b_1)m + 4a_1a_2(c_2 - c_1) + a_2b_1^2 - a_1b_2^2. \end{aligned}$$

Hence there is a common tangent of gradient m if and only if this expression is zero, which is the required condition. Call the left-hand side of that condition $Q(m)$.

Since each root m of $Q(m) = 0$ gives exactly one common tangent, and different roots give different lines (they have different gradients), *the number of common tangents equals the number of distinct real roots of* $Q(m) = 0$. Note also that no common tangent can be vertical, since neither curve has a vertical tangent.



Two parabolas with $a_1 = a_2$ and $b_1 \neq b_2$ have exactly one common tangent; here it has gradient $\frac{1}{2}$.

The case $a_1 \neq a_2$

Now Q is a genuine quadratic in m, so it has exactly one distinct real root if and only if its discriminant is zero. The discriminant is

$$\Delta = 4(a_1b_2 - a_2b_1)^2 - 4(a_2 - a_1)(4a_1a_2(c_2 - c_1) + a_2b_1^2 - a_1b_2^2).$$

The terms without c's simplify nicely:

$$\begin{aligned} &(a_1b_2 - a_2b_1)^2 - (a_2 - a_1)(a_2b_1^2 - a_1b_2^2) \\ &= (a_1^2b_2^2 - 2a_1a_2b_1b_2 + a_2^2b_1^2) - (a_2^2b_1^2 - a_1a_2b_2^2 - a_1a_2b_1^2 + a_1^2b_2^2) \\ &= a_1a_2(b_1^2 - 2b_1b_2 + b_2^2) = a_1a_2(b_1 - b_2)^2. \end{aligned}$$

Therefore

$$\Delta = 4a_1a_2\Big[(b_1 - b_2)^2 - 4(a_1 - a_2)(c_1 - c_2)\Big],$$

and since $a_1a_2 \neq 0$, there is exactly one common tangent if and only if

$$(b_1 - b_2)^2 = 4(a_1 - a_2)(c_1 - c_2). \qquad (*)$$

Ask yourself What exactly does it mean for the two curves to *touch*, and how can we turn that into an equation?

The curves touch when they meet at a point and have the same gradient there. Let

$$d(x) = (a_1 - a_2)x^2 + (b_1 - b_2)x + (c_1 - c_2),$$

the difference of the two quadratics. The curves meet at x_0 when $d(x_0) = 0$, and have equal gradients there when $d'(x_0) = 0$. So they touch if and only if d has a root x_0 with $d'(x_0) = 0$, that is, a repeated root. As $a_1 \neq a_2$, d is a quadratic, and it has a repeated root exactly when its discriminant $(b_1 - b_2)^2 - 4(a_1 - a_2)(c_1 - c_2)$ is zero, which is condition $(*)$.

So, for $a_1 \neq a_2$, the curves have exactly one common tangent if and only if $(*)$ holds, if and only if they touch. (In that case the single common tangent is, naturally, the tangent at the point of contact.)

The case $a_1 = a_2$

Write a for the common value. The m^2 term of Q vanishes and, dividing by $a \neq 0$, the condition becomes

$$2(b_2 - b_1)m + 4a(c_2 - c_1) + (b_1 - b_2)(b_1 + b_2) = 0.$$

- If $b_1 \neq b_2$, this linear equation has exactly one root,

  $$m = \frac{b_1 + b_2}{2} - \frac{2a(c_2 - c_1)}{b_2 - b_1},$$

  so there is exactly one common tangent.
- If $b_1 = b_2$, it reduces to $4a(c_2 - c_1) = 0$. If $c_1 \neq c_2$ there is no common tangent at all (one curve is the other shifted vertically, so their tangents of each gradient are parallel and distinct); if $c_1 = c_2$ the curves are identical and every tangent is common, giving infinitely many.

Hence, when $a_1 = a_2$, the curves have exactly one common tangent if and only if $b_1 \neq b_2$.

Checks

For $y = x^2$ and $y = -x^2 + 2x - \frac{1}{2}$, the difference is $2(x - \frac{1}{2})^2$, so they touch at $x = \frac{1}{2}$; here $Q(m) = -2m^2 + 4m - 2 = -2(m - 1)^2$, with the single root $m = 1$, the gradient of $y = x^2$ at $x = \frac{1}{2}$. For the pictured pair $y = x^2$ and $y = x^2 - 4x + 5$, the formula gives $m = -2 + \frac{5}{2} = \frac{1}{2}$, and both tangent intercepts come out as $-\frac{1}{16}$. We also checked the identity $Q(m) = 4a_1a_2(k_2 - k_1)$ and the discriminant simplification with exact arithmetic on thousands of random coefficient sets.

Common slips

- Claiming that $Q = 0$ having a repeated root "means" the curves touch, without explaining why: the link goes through the *different* quadratic $d(x)$, and the discriminant identity is what connects them.
- In the case $a_1 = a_2$, forgetting the sub-case $b_1 = b_2$, $c_1 = c_2$ (identical curves, infinitely many common tangents), or answering "$b_1 \neq b_2$ or $c_1 \neq c_2$".

Beyond STEP

Describing a curve by its family of tangent lines, rather than its points, is the idea of *duality* in projective geometry: the tangents $y = mx + k$ of a conic satisfy their own quadratic relation between m and k, and two conics in general position share four common tangents, just as they meet in four points. For two parabolas with vertical axes, the line at infinity touches both and uses up two of those four, which is why at most two ordinary common tangents remain. For more problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

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